

Imaadh Usoof

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EDUCATION

University of Melbourne

Bachelor of Science, major in Computing and Software Systems

Melbourne, AUS

Jul. 2024 – Present

PROJECTS

Finance Portfolio Builder | *Python, FastAPI, React, TypeScript, AWS*

January 2026 – Present

- Built a full-stack investment-portfolio web app with a FastAPI backend and a React and TypeScript frontend, serving five years of daily price history and live-updating quotes
- Users can set an investment amount, risk tolerance and time horizon and toggle sector exclusions through a guided, animated interface built for clarity and ease of use
- Deployed to AWS behind nginx with automatic HTTPS, running the API as a background service with a nightly data-refresh job; live at finance.imaadhusoof.com

PDF annotation app | *Electron, React, TypeScript, SQLite*

March 2025 – June 2025

- Built an application for lecture note taking and PDF annotation using Electron and TypeScript
- Implemented stylus and mouse-based annotation tools, used SQLite to store offline edits
- Ensured app has crucial PDF editing features only found on paid alternatives such as Adobe Acrobat

Space shooter | *Python, pygame, TensorFlow*

May 2019 – August 2019

- Built an endless space shooter game in Python using pygame, featuring asteroid waves, boss battles, and WASD controls
- Trained a reinforcement learning agent using TensorFlow to autonomously play the game, designing the game's control scheme (left/right only) to make it learnable by a simple neural network

Spellchecker Dictionary | *C, Patricia Trie* | Coursework

September 2025 – October 2025

- Implemented a spellchecker in C using a Patricia trie (compressed radix tree), supporting exact and fuzzy key lookup across a large address dataset
- Engineered bit-level string operations including custom getBit and createStem functions to navigate the trie using binary representations of characters
- Applied edit distance algorithms for closest-match recommendations when no exact key is found, traversing all descendant nodes at the mismatch point
- Conducted a formal complexity analysis comparing the Patricia trie against a linked list implementation, varying input size and key characteristics to validate Big-O behaviour empirically

TECHNICAL SKILLS

Relevant Coursework: Real Analysis, Probability, Algorithms and Data Structures, Machine Learning

Languages: Java, Python, C, SQL, JavaScript

Frameworks: FastAPI, React, Electron, SQLAlchemy

Databases: MySQL, PostgreSQL, Amazon RDS

Libraries: Tensorflow, pandas, NumPy, Matplotlib, pygame, PyTorch

Portfolio: imaadhusoof.com